PROJECT NAME: **Vaccine Integrity Monitoring System (VIMS)**

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TASK #	PROJECT TASKS WHICH REQUIRE SOFTWARE TOOL SUPPORT
1	IoT Data Acquisition & Transmission
2	Database
3	AI-Powered Predictive Analytics Engine
4	Real-Time Monitoring Platform

HARDWARE TOOLS FOR TASK 1: IoT Data Acquisition & Transmission

Tool Cost/Training/Functionality Data

Tool	Arduino Portenta H7	Arduino Nano ESP32	Arduino Mega 2560	Raspberry Pi Zero 2 W
Cost	$120	$20	$25	$20
Training Days	7 Days	3 Days	2 Days	6 Days
Functionality	90	85	80	60

Normalized Cost/Training/Functionality Data (Higher is better)

Tool	Arduino Portenta H7	Arduino Nano ESP32	Arduino Mega 2560	Raspberry Pi Zero 2 W
Cost	**100**	16.7	20.8	16.7
Training Days	**100**	42.9	28.6	85.7
Functionality	**100**	94.4	88.9	66.7

Normalized Tool Graph

Which tool has been selected? Why?

Selected Tool: Arduino Nano ESP32

Arduino Nano ESP32 was selected because it offers the best balance between cost, training time, and functionality. It includes built-in Wi-Fi and Bluetooth, making it highly suitable for IoT-based data acquisition and transmission.Compared to the other options, it is more cost-effective than Arduino Portenta H7, more modern and communication-ready than Arduino Mega 2560, and easier to use for embedded sensor control than Raspberry Pi Zero 2 W. Overall, it is the most practical and efficient choice for this project.

SOFTWARE TOOLS FOR TASK 2: Database

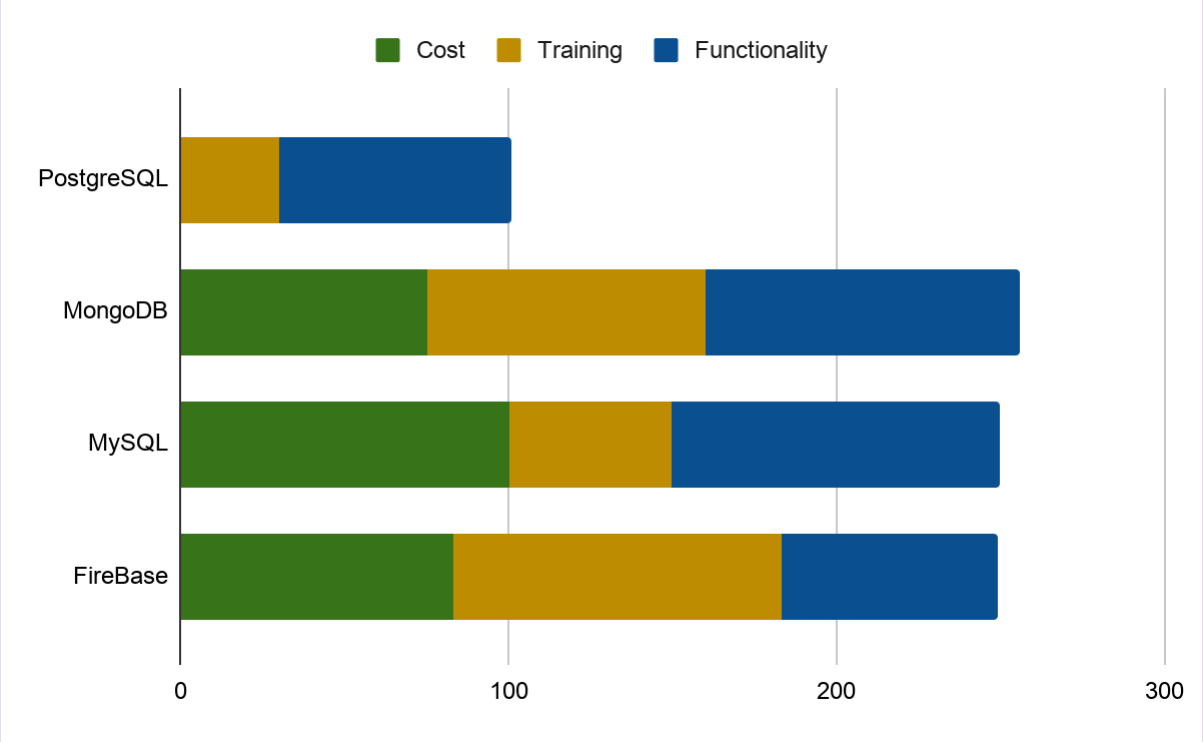
Tool Cost/Training/Functionality Data

Tool	PostgreSQL	MongoDB	MySQL	Firebase
Cost	$0	$0	$0	$25
Training Days	5 Days	4 Days	3 Days	2 Day
Functionality	95	90	85	80

Normalized Cost/Training/Functionality Data

Tool	PostgreSQL	MongoDB	MySQL	Firebase
Cost	**0**	0	0	100
Training Days	100	80	60	**40**
Functionality	100	94	**89**	84

Normalized Tool Graph



Which tool has been selected? Why?

Selected Tool: MongoDB

MongoDB was selected because it offers a high functionality score and a balanced training period, making it ideal for managing the continuous data flow from IoT sensors. Its document-oriented structure allows for the seamless storage of real-time temperature, humidity, and movement logs without the rigid schema constraints of traditional databases. Compared to PostgreSQL, it provides more flexibility for rapid prototyping of sensor data; compared to MySQL, it handles unstructured AI-driven datasets more efficiently; and it is significantly more cost-effective for large-scale data than Firebase. This makes it the most scalable and efficient backend choice for ensuring the integrity of the vaccine cold chain.

SOFTWARE TOOLS FOR TASK 3: AI-Powered Predictive Analytics Engine

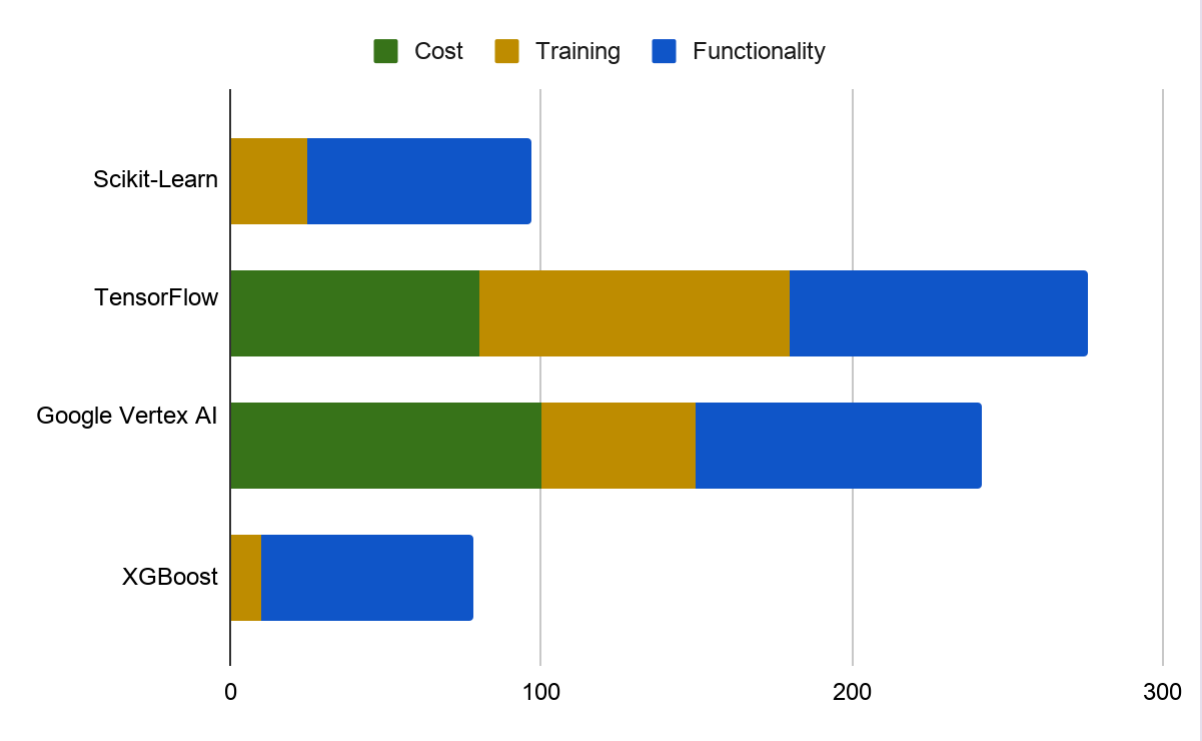
Tool Cost/Training/Functionality Data

Tool	Scikit-Learn	TensorFlow	Google Vertex AI	XGBoost
Cost	$0	$0	$50	$0
Training Days	4 Days	7 Days	6 Days	3 Days
Functionality	85	95	100	90

Normalized Cost/Training/Functionality Data

Tool	Scikit-Learn	TensorFlow	Google Vertex AI	XGBoost
Cost	**0**	0	100	**0**
Training Days	57	**100**	85	42
Functionality	85	**95**	100	90

Normalized Tool Graph



Which tool has been selected? Why?

Selected Tool: TensorFlow

TensorFlow was selected because it offers the highest functionality among open-source options, making it the most capable tool for complex predictive analytics. While it requires a longer training period compared to other frameworks, its ability to handle large-scale datasets and support advanced deep learning architectures is essential for the AI-Powered Predictive Alerting objective. Unlike Scikit-Learn or XGBoost, TensorFlow provides superior flexibility for building custom neural networks that can analyze intricate environmental trends and movement patterns in real-time. By prioritizing long-term functionality and scalability over short-term training speed, TensorFlow ensures the system can achieve the 15-second predictive notification goal with high precision.

SOFTWARE TOOLS FOR TASK 4: Real-Time Monitoring Platform

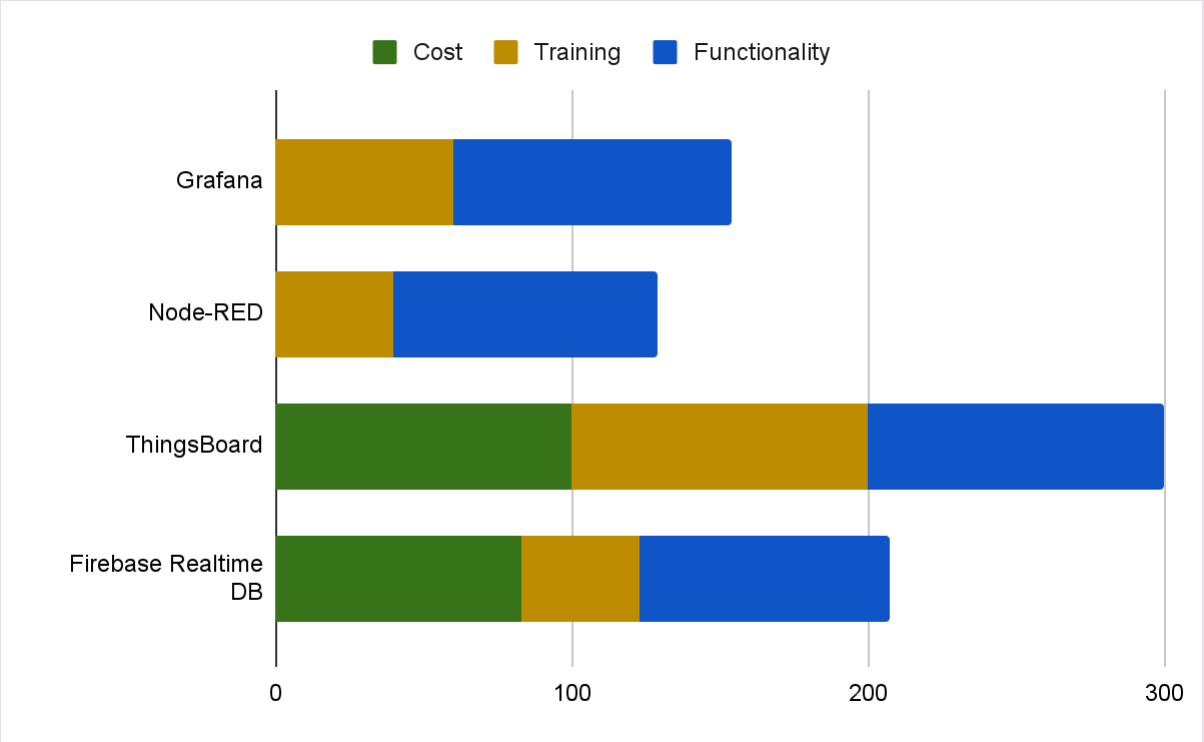
Tool Cost/Training/Functionality Data

Tool	Grafana	Node-RED	ThingsBoard	Firebase Realtime DB
Cost	0	0	30	25
Training Days	3	2	5	2
Functionality	90	85	95	80

Normalized Cost/Training/Functionality Data

Tool	Grafana	Node-RED	ThingsBoard	Firebase Realtime DB
Cost	**0**	0	100	**83**
Training Days	60	**40**	100	40
Functionality	94	**89**	100	84

Normalized Tool Graph



Which tool has been selected? Why?

Selected Tool: Grafana

Grafana was selected because it provides an outstanding balance between high functionality (94/100) and low cost ($0), making it the most efficient choice for a Real-Time Monitoring Platform. While ThingsBoard offers slightly higher functionality, Grafana delivers nearly the same capability with significantly lower training requirements and zero licensing costs. Its robust ability to visualize time-series data makes it perfect for tracking the 60-second temperature and humidity intervals required for vaccine safety. Compared to Node-RED or Firebase, Grafana offers superior dashboarding and alerting features out of the box, ensuring the system can reliably meet the 99% uptime and real-time visualization objectives of the VIMS project.